A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, I
Revision Log: Draft 9 to Draft 10 (exposition only)

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Scope

A follow-up to the exposition pass of Draft 9, on David Joyner's replies to the points raised in the Draft 9 log. **No mathematical change**: no statement, proof, hypothesis or label was changed. All section, theorem, equation and appendix numbers are the same as in Drafts 8 and 9 (checked by comparing the `.aux` label tables).

Decisions recorded

1. **[O] means "proof outlined"** (David). This is the wording already used in the papers, the README and `glossary.tex`; nothing changed.

2. **The glossary stays at the back** (§10) (David). Nothing changed.

Changes

1. **Klose's work summarized in the literature review** (§11, paragraph "Central extensions of division algebras"), at David's request, with emphasis on *Metaplektische Erweiterungen von Quaternionenschiefkörpern* (Math. Z. 193). The new text covers: the dissertation (Göttingen 1983, advisor S. J. Patterson) and the two papers drawn from it; the subject of the local paper (Math. Z. 191); the global paper (Math. Z. 193): explicit cocycles and partial results on topological central extensions of the adele group of $\mathrm{GL}_{1,D}$ for a quaternion division algebra D; the facts on commutators and symmetric cocycles attributed to Klose by Hill, including $H^2_{\mathrm{sym}}(\mu_m, \mu_m) \cong \mathbb{Z}/m$; the use of Klose's result in [J] to choose a quaternion D for which the cover of $G_D(\mathbb{A}_K)$ exists without $(n, r) = 1$; and the difference from the construction of the cover in [JC]. The old sentence "we have not compared it with Klose's construction" is replaced by an explicit statement of what was and was not consulted.

   Sources and their limits. We could not obtain the text of either Klose paper or of the dissertation: the Springer page and the Göttingen digitization (GDZ) were not reachable from this session. The summary therefore rests on:

   - the EuDML records: Math. Z. 193, keywords "central extensions; quaternion algebra; constructions of cocycles", MSC 20G10, 20G30, 11R52, 16Kxx; Math. Z. 191, keywords

"commutators; two-cohomology class; central extensions; quaternionic skew field; metaplectic extensions; structure of; one-units", MSC 11S45, 16U60, 16Exx, 16Kxx;

- Prasad–Rapinchuk, Publ. IHÉS 84 (1996), Introduction (read on numdam), which describes Klose's results as partial results on topological central extensions of the adele group associated with $\mathrm{GL}_{1,D}$ for D a quaternion division algebra;
- R. Hill, arXiv:0708.0108, §2.3 ("Commutators and symmetric cocycles"), which says its facts were used in its reference [16] to determine the metaplectic extensions of $D^\times$ for a quaternion algebra over a global field, and §3.8, which calls the isomorphism $H^2_{\mathrm{sym}}(\mu_m, \mu_m) \cong \mathbb{Z}/m$ "Klose's isomorphism". We could not see Hill's bibliography, so the identification of his [16] with one particular Klose paper is not confirmed; the review attributes the facts to Klose, as Hill does, without naming which paper;
- David Joyner's 1988 paper [J], §2.6, as posted on yetanothermathblog.com (it cites Klose's dissertation);
- Springer's article page (pages 625–649, received October 8, 1985; it cites [KP1] and Klose's Math. Z. 191 paper) and the Mathematics Genealogy Project (dissertation title and advisor).

The review says in the text that the summary rests on these and that Klose's theorem statements are not reproduced. If you can obtain the Math. Z. 193 paper (or the dissertation), a closer summary, and a comparison with the cover of [JC] for $r = 2$, would be worth making.

2. **Bibliography.** [Klo86b] pages corrected to 625–649, as on the publisher's page (EuDML and GDZ list 625–650; Prasad–Rapinchuk also give 625–649). New entry [Hil] (R. Hill, arXiv:0708.0108; author and title checked on arXiv).

3. Title line: Draft 10. Date unchanged (November 8, 2026).

Still open from the Draft 9 log

1. Flicker–Kazhdan, "A simple trace formula" (1988) versus Proposition 6.2: no decision yet.

2. The companion paper: its literature section's sentence that we are "not aware of an independent literature on n-fold covers of division algebras beyond [JC]" should now point to Klose, and its Moore entry should read 5–70. To be done when the companion paper is next edited.

3. Subsection lead-ins: kept as in Draft 9 unless you want them pruned.

Files

`part-I/mjl_part1_draft10.tex` and `.pdf`; this log (`.tex`, `.pdf`); `README.md` (Part I draft number). Drafts 8 and 9 are kept unchanged; `glossary.tex` is unchanged.